Hamza Ahmed Siddiqui

Data Scientist · ML Engineer · AI Solutions

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Work Experience

Research Assistant - Computer Vision **Apr 2026 – Present**
Eckerlab *Göttingen, Germany*
Project: Video Generalization Dataset (VGD)

- Conducted an empirical research study with **hypothesis-driven experiments** to quantify object bias in **video action recognition models**, benchmarking Something-Something-v2 dataset against a custom-curated, multi-camera distractor dataset.
- Quantified fine-grained object bias in action recognition datasets by **engineering a multi-camera synchronized capture system** (custom 3-camera hardware setup) to record controlled validation data.
- Architected multi-modal feature extraction and cross-attention probing pipelines for frozen **vision foundation models** (DINOv2, V-JEPA2), **optimizing I/O throughput, GPU utilization and dynamic batching**.
- Developed an automated object-occlusion pipeline utilizing SAM3 to isolate and mask bounding boxes across ~150k frames, proving spatial object reliance in single-frame baselines.
- Evaluated a 320B-parameter Vision-Language-Model (glm-5.3-flash) for object bias via **cloud APIs, engineering structured JSON prompts** with base64 video representations over a condensed label space.

Project: Enhancing Domain Generalization in Video Action Recognition

- Trained two-stream video classification architectures (I3D ConvNets) to evaluate multi-modal inputs (RGB and Optical Flow) for mitigating background bias in **cross-domain generalization**.
- Built Optical Flow **extraction and caching** pipeline with Flow visualization option.
- Built modular **PyTorch training pipelines from scratch** incorporating WandB experiment tracking, early stopping, LR scheduling, and dynamic model checkpointing.
- Optimized data loader throughput by serializing multi-modal datasets into HDF5 (.h5), significantly cutting I/O bottleneck and **maximizing GPU utilization**.
- Designed a late-fusion inference strategy combining RGB and flow modalities with hyperparameter optimization, **improving uAP from 72.8 to 75.2** on heavily imbalanced dataset.

Teaching Assistant - NLP **Oct 2025 – Present**
GippLab *Göttingen,Germany*
Project: Teaching Assistant for courses:

- Deep Learning for NLP, Programming for Data Science

Senior Data Scientist | Data Scientist | Junior Data Scientist **Jun 2020 – Apr 2025**
Afiniti *Istanbul, Türkiye*
Afiniti uses patented AI to pair participants in customer interactions in a way that optimizes business goals like revenue.

- Maintained and optimized core **Python and R pipelines utilizing Stan to perform Bayesian estimation** of agent and caller quality from high-volume historical call center logs.
- Engineered production data integrity and automated drift-detection systems in Python, R and SQL, reducing silent pipeline failures and safeguarding model reliability across real-time customer pairing systems.
- Architected production-ready ML and data processing pipelines using Python, SQL, and Apache Spark, optimizing ETL throughput for **multi-million record customer datasets** across cloud environments.
- Owned **end-to-end model shipment** by designing feature engineering workflows, conducting **reproducible validation experiments**, and **deploying models** directly into production.
- Designed and executed time-cycle **A/B testing** frameworks and causal inference models for enterprise customer-agent pairing, driving a **25% incremental revenue** lift over baseline FIFO routing.
- Improved next-purchase propensity prediction by increasing ROC-AUC from **0.75** to **0.85**, by training LightGBM models on binary sale outcomes.
- Enabled **13%** sales uplift in AI-agent pairing, by integrating customer purchase-probability scores into the matching platform to prioritize customers based on predicted likelihood of conversion.

Leadership experience:

- Led the transformation of a loss-making business unit into a **$300,000/month** profit making unit.
- Led a team of 6 data scientists; mentored new hires and presented in stakeholder meetings.

Education

MSc in Applied Data Science **Apr 2025 – Present**
Georg-August-Universität Göttingen *Germany*
Thesis: "TBD"
Overall Grade: ***1.2***

BSc Economics **Aug 2016 – May 2020**
Lahore University of Management Sciences (LUMS) *Lahore, Pakistan*
Overall Grade: ***3.91****/4.00 (High Distinction)*

Selected Projects

Berlin Airbnb Price Prediction Project **Jul 2025**
ML model to predict Airbnb listing prices using features such as location, property type, amenities, and historical pricing data.

- Implemented Catboost Regressor and optimized performance using hyperparameter tuning, achieving strong test accuracy of ~85%. [Github]

ResumeReviewer-Agent **Oct 2026**
Built an agentic AI resume analyzer using **LangChain (LCEL)**, Gemini, and Streamlit to automate candidate skill gap analysis against target job descriptions.

- Engineered a document parsing pipeline (PyMuPDF/Unstructured) and used **Pydantic** structured JSON outputs to generate strictly typed, actionable resume rewrite suggestions. [Github]

More projects on my webpage: https://hamzasiddiqui10.github.io/

Technical Skills

- **Programming & Data:** Python, R, Java, Scala, C/C++, SQL, PostgreSQL, MongoDB, Redis, Neo4j, InfluxDB,
- **Backend & Pipelines:** Linux, FastAPI, Kafka, Spark, Airflow, Docker, Kubernetes, CI/CD, GitHub Actions, Conda, Slurm, HDF5 / h5py.
- **Cloud:** Azure Functions, Azure Blob Storage, Azure VMs, Azure PostgreSQL, Azure Key Vault, AWS S3, Hadoop, Spark, Databricks
- **Methodologies:** Machine Learning, Statistical Modeling, Prompt Engineering, Agentic AI, AI-Driven Code Generation.
- **ML & Analytics:** PyTorch, Pandas, Scikit-Learn, HuggingFace, LoRA, A/B testing, statistical analysis, Neural Networks, Transformers, Regressions, Decision Trees, Ensemble Models (Random Forest, XGBoost, LightGBM, CatBoost), Bayesian Methods, PCA, GMMs, t-SNE, Feature Engineering, Hyperparameter Tuning, ML workflow, Langchain, Langsmith, PyDantic, Gemini API, openai API, AdamW, LR scheduling.
- **Computer Vision & Video Processing:** PyAV, Decord, DINOv2, V-JEPA / V-JEPA 2, SAM3 (Segment Anything Model), Vision-Language Models (VLMs), I3D, ViTs.
- **NLP:** LLMs, Prompt Engineering, Structured LLM Output, Retrieval-Augmented Generation (RAG), Information Retrieval, Embeddings and Representational Learning, Knowledge Distillation, BERT.

Honors & Awards

- LUMS Dean's Honor List, 8 out of 8 semesters
- LUMS Graduation with High Distinction (Top 1% in university)
- 100% Scholarship in GCE Advanced Level (High School): Based on excellent grades in O Level

Language Proficiency

English: Native **German:** Intermediate

Other Interests

Football, Tennis, Swimming, Watches, History